THE Z/GR CONSTRUCTION INSTANTIATED ON A PROPER SCHEME WITH AN AMPLE LINE BUNDLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 14491-14512 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 14514-14520, the complete author proof (5 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Application card of the chapter's Z-graded-geometry section feeding the weight/picture arguments (L16088's neighbour).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Morphisms, lemma proper-ample-is-proj; Properties, lemma ample-gcd-is-one; the chapter's Proj/Spec diagram construction (equation-proj-and-spec) whose properties are imported wholesale.

Source reference. Original lemma 0EKG: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14398–14412. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0C6J: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14520–14520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 01QI: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 14520–14520. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let R be a ring. Let P be a proper scheme over R and let $\mathcal{L}$ be an ample invertible $\mathcal{O}_P$-module. Set $A = \bigoplus_{m \geq 0} \Gamma(P, \mathcal{L}^{\otimes m})$. Then $P = Proj(A)$ and diagram (0EKG) becomes the diagram*

$$\begin{array}{ccccc} \underline{\mathrm{Spec}}_P\left(\bigoplus_{m\in\mathbf{Z}} \mathcal{L}^{\otimes m}\right) & \longrightarrow & L = \underline{\mathrm{Spec}}_P\left(\bigoplus_{m\geq 0} \mathcal{L}^{\otimes m}\right) & \xrightarrow{\pi} & P \\ \| & & \downarrow{\scriptstyle \sigma} & & \downarrow \\ U & \longrightarrow & X & \longrightarrow & Z \end{array}$$

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having the properties explained above.

Proof. We have $P = \mathrm{Proj}(A)$ by Morphisms, Lemma 0C6J. Moreover, by Properties, Lemma 01QI via this identification we have $\mathcal{O}_P(m) = \mathcal{L}^{\otimes m}$ for all $m \in \mathbf{Z}$. $\square$